

Advanced Software Development

Software Development Fundamentals



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Software Development Fundamentals

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- VCS Hosting
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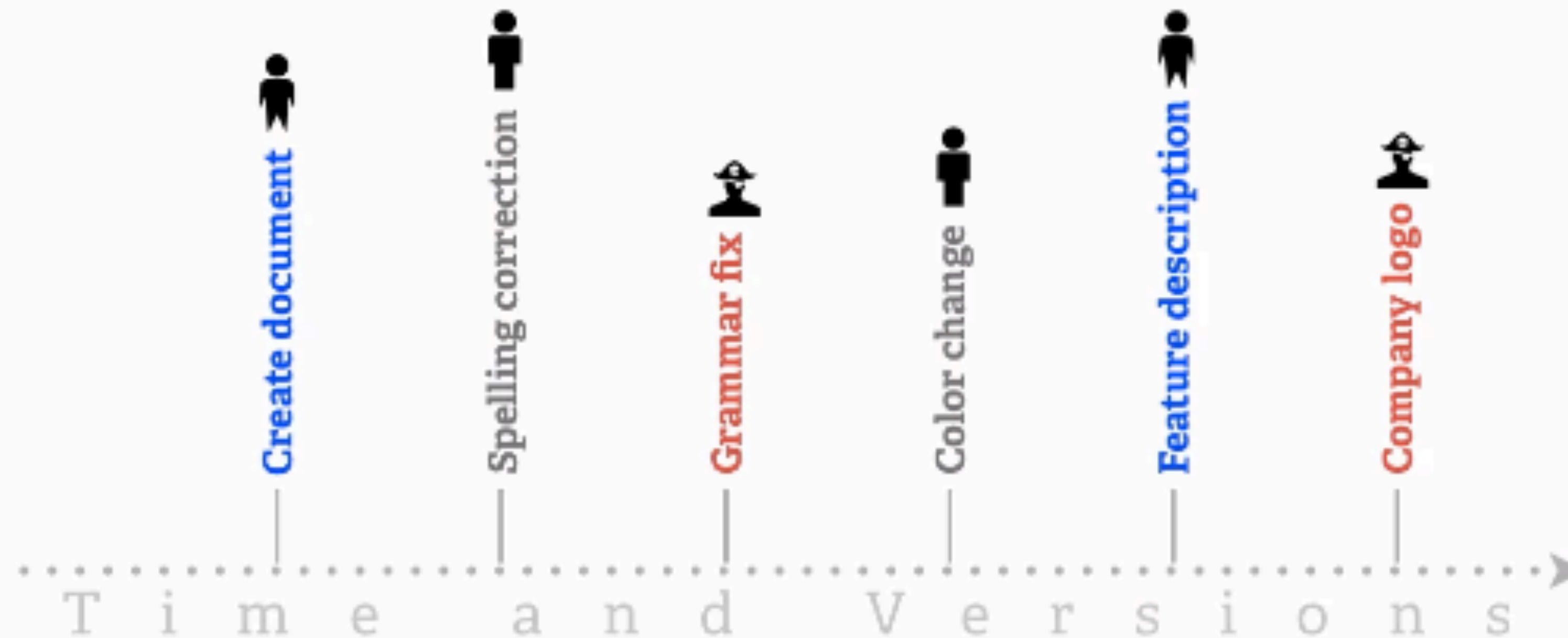
Version Control Systems

- tools that automate tracking, managing, and storing changes to computer files.
- VCS types:
 - Distributed: e.g., Git, Mercurial, Bazaar
 - Full copy of the project is kept local.
 - Centralised: e.g., Apache Subversion (SVN), Perforce Helix Core, CVS.
 - Tracking is done from a central location.
 - Offers better administrative control
 - Lock-based: e.g., DBMS

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Local Version Control System

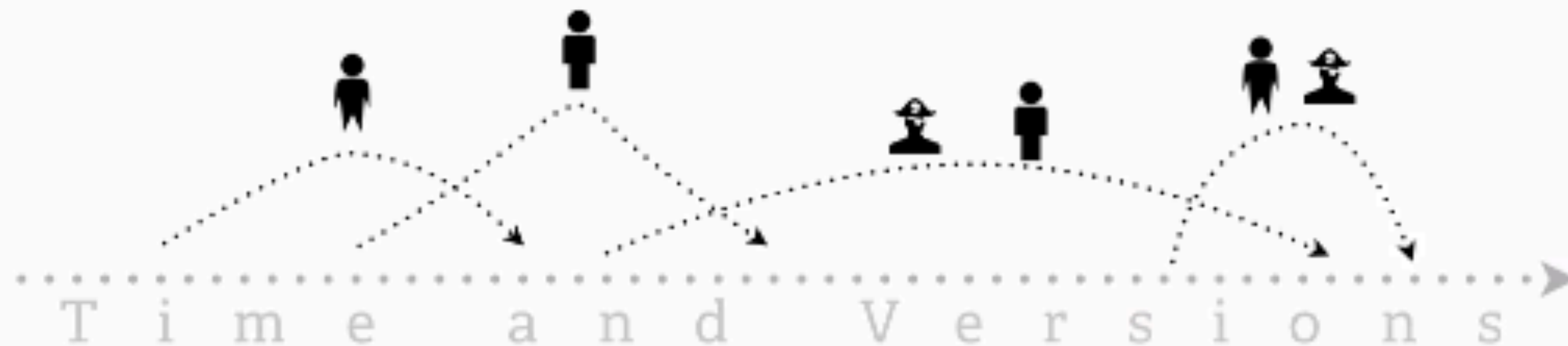
History Tracking



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Collaborative Version Control System

Collaborative History Tracking



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Core Concepts

- Repo: the central repository for all project files.
- Commit: changes made, usually accompanied by a unique ID and a message describing the update.
- Branching: creating a separate file to develop features or fix bugs without affecting the main project.
- Merging: combining changes from different branches into one, often involving conflict resolution.
- Traceability: ability to see who, when, and why changes were made.

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Git Common Actions

- `git init myproject` [sets up or initialise a project called myproject]
- `cd myproject` [myproject]
- `git add .`
- `git commit -m "Importing all the code"`
- `git checkout master`
- `git commit -a -m "My new logo"`
- `git push`

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Why Git

- fast and modern way of implementing version control.
- provides a history of content changes.
- facilitates collaborative changes to files.
- easy to use for any type of projects by:
 - programmers.
 - designers.
 - document authors.
 - ...

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Git Version Control Best Practices

- make incremental, small changes:
 - reduces merge conflicts, makes testing easier, and simplifies error rollback.
- keep commits atomic.
- develop using branches:
 - keeps unstable code from production, helping teams test safely and collaborate more effectively.
- write descriptive commit messages.
- obtain feedback through code reviews.

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Further Study

- Git codelab:

<https://toolkit.codelabdavis.com/developers/overview/version-control/>

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References

- Git Version Control:

<https://about.gitlab.com/topics/version-control/what-is-git-version-control/>

- Git Cheat Sheet:

<https://git-scm.com/cheat-sheet>

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Thank You

Questions?

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